Unit II :: Mathematical Typesetting and Technical Elements in LaTeX

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Syllabus

Unit II: Mathematical Typesetting and Technical Elements in LaTeX

Math mode: inline and display modes; math fonts: mathcal, mathbb, mathds, mathfrak; superscripts and subscripts, pre-defined functions, fractions, roots, Greek letters; integrals, summations and products; mathematical grids: multi-line formulae, gathering system of equations, aligning system of equations, matrices; LaTeX packages for Physics: siunitx, physics; floats in LaTeX: table and figure environments; typesetting tables using tabular; formatting figures using includegraphics.

1 Overview of math mode in LaTeX

In LaTeX, the typesetting of mathematical expressions requires entering a dedicated operating environment known as *math mode*. In math mode, LaTeX does the following

- Treats alphabet symbols as mathematical variables.
- Typesets letters in italics by default.
- Ignores standard whitespace characters.
- Adjusts spacing between symbols dynamically based on their semantic classification, such as operators, relations and delimiters.

1.1 Inline math mode

Inline math mode is used to embed short mathematical expressions and variables directly within a sentence without disrupting the paragraph flow. We can typeset a mathematical expression in inline mode using any of the following syntax:

1. Enclosing the expression within `$...$`, for example `$F=ma$`.
2. Enclosing the expression within `\(...\)`, for example `\(p=mv\)`.
3. Placing the expression within the `math` environment, for example
```
\begin{math}
    U=mgh
\end{math}
```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $F=
ma$, linear momentum \(p=mv\), and gravitational potential energy
\begin{math}
  U=mgh
\end{math}.
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion $F = ma$, linear momentum $p = mv$, and gravitational potential energy $U = mgh$.

1.2 Display math mode

Display math mode is used for prominent, complex or multi-line equations that require dedicated vertical separation, centered alignment and full symbol scaling.

For display mode, we have two types of syntax:

1. *Un-numbered display syntax*:
   (a) Enclosing the expression within `$$...$$`, for example `$$F=ma$$`.
   (b) Enclosing the expression within `\[...\]`, for example `\[p=mv\]`.
   (c) Placing the expression within the `displaymath` environment, for example
   ```
   \begin{displaymath}
       U=mgh
   \end{displaymath}
   ```
2. *Numbered display syntax*: Placing the expression within the `equation` environment, in which case the equation will also be given a number. For example
   ```
   \begin{equation}
       T=\frac{1}{2}mv^2
   \end{equation}
   ```

For instance, the following code contains all of the above three syntax

```
In classical mechanics, an object's translational motion and energetic state
near Earth's surface can be characterized by Newton's second law of motion $$F=
ma$$ linear momentum \[p=mv\] gravitational potential energy
\begin{displaymath}
  U=mgh
\end{displaymath}
and kinetic energy
\begin{equation}
  T=\frac{1}{2}mv^2
\end{equation}
```

which produces the following output:

In classical mechanics, an object's translational motion and energetic state near Earth's surface can be characterized by Newton's second law of motion

$$F = ma$$

linear momentum

$$p = mv$$

gravitational potential energy

$$U = mgh$$

and kinetic energy

$$T = \frac{1}{2}mv^2 \tag{1}$$

2 Mathematical typography in LaTeX

2.1 Superscripts and subscripts

Superscripts and subscripts are commonly used to represent exponents (such as 3^2, x^{-3}) and indexed quantities (such as m_1, K_0) respectively. In LaTeX, superscripts are typeset using the *caret* symbol (`^`) while subscripts are typeset using the *underscore* symbol (`_`).

If a superscript or subscript contains more than a single character, the entire exponent or index must be enclosed in braces `{ }`.

For instance, the following code contains expressions with exponents and indices

```
$$ax^2+bx+c=0$$

$$f(x)=e^{-ax^2}$$

$$m_1=5, m_{12}=36.5$$
```

which produces the output

$$a x^2 + b x + c = 0$$

$$f(x) = e^{-a x^2}$$

$$m_1 = 5, m_{12} = 36.5$$

2.2 Fractions and roots

In LaTeX, fractions are typeset using the `\frac` macro. The general syntax is

```
\frac{numerator}{denominator}
```

For fractions contained inline, where vertical space is limited, standard slash notation `a/b` or the `\tfrac{num}{den}` macro can be used.

For instance, the following code contains examples of typesetting fractions

```
The kinetic energy of a particle of mass $m$ and moving with speed $v$ is $\
frac{1}{2}mv^2$.

Using the law of equipartition of energy, the average energy of a monoatomic
molecule is obtained to be $\tfrac{3}{2}k_BT$.
```

which produces the output

The kinetic energy of a particle of mass m and moving with speed v is $\frac{1}{2}mv^2$.
Using the law of equipartition of energy, the average energy of a monoatomic molecule is obtained to be $\frac{3}{2}k_BT$.

LaTeX allows us to typeset the square root using the `\sqrt` macro which has the general syntax as

```
\sqrt{expression}
```

The `\sqrt` macro can be extended to typeset the nth root using an optional argument. The general syntax for this is

```
\sqrt[n]{expression}
```

For instance, the following code contains examples of typesetting roots

```
The root mean squared speed of molecules is $\sqrt{\dfrac{3k_BT}{m}}$, as per
Maxwell-Boltzmann distribution.

The roots of a quadratic equation are
\begin{equation}
  x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}
\end{equation}
```

which produces the output

The root mean squared speed of molecules is $\sqrt{\dfrac{3k_BT}{m}}$, as per Maxwell-Boltzmann distribution.
The roots of a quadratic equation are

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a} \tag{2}$$

2.3 Pre-defined functions and operators

Standard mathematical functions (such as trigonometric, logarithmic and limit operators) must not be typed as plain text in math mode; otherwise, LaTeX would render them in variable italics. For example, `sin(x)` typesets as a product of s, i and n. Hence to render this correctly in upright roman typeface with appropriate surrounding whitespace, LaTeX provides pre-defined operator macros for a bunch of functions as given below.

- *Trigonometric and hyperbolic functions*: `\sin`, `\cos`, `\tan`, `\arcsin`, `\arccos`, `\arctan`, `\sinh`, `\cosh`, `\tanh`
- *Logarithmic and exponential functions*: `\exp`, `\log`, `\ln`
- *Limits and optimization*: `\lim`, `\min`, `\max`, `\sup`, `\inf`
- *Linear algebra*: `\det`, `\ker`, `\deg`

2.4 Summation, product and integral operators

In LaTeX, the summation, product and integral operators are typeset using the \sum, \prod and \int macros. These are variable-sized operators and get rendered in different sizes, depending on whether they are in the inline or display modes.

The following code typesets these in the inline mode:

```
$\sum_{i=1}^{N}(2i-1)$, $\prod_{k=1}^{N}\frac{1}{2^k}$ and $\int_{0}^{5}e^{-x}
dx$
```

which gives the following output:

$\sum_{i=1}^{N}(2i-1)$, $\prod_{k=1}^{N}\frac{1}{2^k}$ and $\int_{0}^{5}e^{-x}dx$

In display math mode, variable-sized operators render larger than in inline mode, and their limits adjust automatically. The following code typesets the above expressions in the display mode:

```
$$\sum_{i=1}^{N}(2i-1)$$ $$\prod_{k=1}^{N}\frac{1}{2^k}$$ $$\int_{0}^{5}e^{-x}
dx$$
```

which gives the following output:

$$\sum_{i=1}^{N}(2i-1)$$

$$\prod_{k=1}^{N}\frac{1}{2^k}$$

$$\int_{0}^{5}e^{-x}dx$$

2.5 Greek letters

In physics, letters from the Greek alphabet are widely used to represent various physical quantities. In LaTeX, lowercase Greek letters are typeset by prefixing the letter's English name with a backslash, as given in the following macros:

Letter	Macro	Letter	Macro	Letter	Macro	Letter	Macro
α	\alpha	η	\eta	ν	\nu	υ	\upsilon
β	\beta	θ	\theta	ξ	\xi	ϕ	\phi
γ	\gamma	ι	\iota	π	\pi	χ	\chi
δ	\delta	κ	\kappa	ρ	\rho	ψ	\psi
ϵ	\epsilon	λ	\lambda	σ	\sigma	ω	\omega
ζ	\zeta	μ	\mu	τ	\tau		

2.6 Mathematical fonts